

FEDERAL UNIVERSITY OF TECHNOLOGY AKURE

SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY

DEPARTMENT OF INFORMATION AND COMMUNICATION ENGINEERING

SECOND SEMESTER 2022/2023 ACADEMIC SESSION EXAMINATION

COURSE: DATABASE MANAGEMENT SYSTEM

CODE: ICT 202

TIME: 2 HOURS

INSTRUCTION: ANSWER QUESTION ONE AND ANY OTHER TWO QUESTIONS

Question 1 ✓

a. State Armstrong inference rule and list the six functional dependency (FD) inference rules (10 marks)

1b. Let $R = (A, B, C)$, $S = (C, D, E)$ and let q and r be the relation on Schema R , and s be the relation on Schema S . convert the following Queries to SQL

(i) $\{ \langle a \rangle \mid \exists b (\langle a, b \rangle \in r \wedge b = 10) \}$ = Select ^{list of} ~~a~~ from r when $b = 10$

(ii) $q - r$

(iii) $\{ t \mid \exists p \in r \exists q \in S(t) [A] - p[A] \wedge t[E] - q[E] \wedge p[C] - q[D] \}$ (10 marks)

c. For the following relation Schema:

Employee (employee-name, street, city)

Works (employee-name, company-name, salary)

Company (company-name, city)

Manages (employee-name, manager-name)

Give an expression using appropriate SQL COMMAND for each of the following Queries

(i). Find the names, street address, and cities of residence for all employees who work for first bank corporation and earns N10,000, (2 marks)

(ii). Find the names of all employees in the database who live in the same city as the company for which the work (2 marks)

(iii). Find the names of all employees in the database who live in the same cities as their managers (2 marks)

(iv). Find the names of all employees in the database who do not work for first bank corporation. Assume that people work exactly in one company (2 marks)

(v). Find the names of all employees in the database who earn more than every employee of small bank corporation. Assume that people work for at most one company. (2 marks)

Question 2 ✓

a. Briefly describe the following categories of DBMS data models: (a) Relational model, (b) Entity relationship model, (c) Object based model, and (d) semi structured model, (5 marks)

- b. Enumerate five major advantages of Database management system over old file system (5 marks)
- c. Justify the need for data abstraction and what is data abstraction levels? (5 marks)

Question 3

- a. Briefly describe the main features of SQL in database management system (5 marks)
- b. Clearly distinguish (with sketches where applicable) the differences between a database SCHEMA and database INSTANCE (5 marks)
- c. What is Query in SQL? (5 marks)

Question 4 ✓

- a. Differentiate between Primary keys, Surrogates keys and Foreign keys (5 marks)
- b. Write short notes with symbol sketches where applicable on (a) Attributes, (b) Multivalued attributes, (c) Entity instance, (d) weak and strong entities (5 marks)
- c. With detailed explanations distinguished between the 5 SQL commands viz DDL, DML, DCL, TCL, DQL (5 marks)

Question 5 ✓

- a. Justify the need for data Normalization in DBMS (5 marks)
- b. Explain the three data modification anomalies. (5 marks)
- c. Write short notes on the following Normal Forms (i) 1NF, (ii) 2NF, (iii) 3NF, and (iv) BCNF (5 marks)